

Imbalanced Mach-Zehnder Interferometers for Group-Index Extraction

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1. Objective

Measure the group index n_g of the standard 500×220 nm silicon strip waveguide (TE, 1550 nm) from the free spectral range (FSR) of three imbalanced Mach-Zehnder interferometers (MZIs) with different path-length differences ΔL , and compare it with the value predicted by an eigenmode (FDE) compact model. A grating-coupler loopback is included to de-embed the grating-coupler (GC) spectral envelope.

2. Theory and design equations

With ideal Y-branches (50/50, in-phase outputs) the MZI transmission is

$$T(\lambda) = \frac{1}{2} [1 + \cos(\beta \Delta L)], \quad \beta = 2\pi n_{\text{eff}}(\lambda) / \lambda$$

Adjacent transmission peaks differ in phase by 2π , which gives

$$\text{FSR} = \lambda^2 / (n_g \Delta L), \quad n_g = n_{\text{eff}} - \lambda \, \text{d}n_{\text{eff}}/\text{d}\lambda$$

A Lumerical MODE (FDE) wavelength sweep of the 500×220 nm strip (Si and SiO₂ Palik) was fitted to a compact model $n_{\text{eff}}(\lambda) = 2.4468 - 1.1334 (\lambda - 1.55) - 0.0439 (\lambda - 1.55)^2$ (λ in μm), giving **$n_g = 4.204$ at 1550 nm**. Measuring the FSR therefore yields $n_g = \lambda^2 / (\text{FSR} \cdot \Delta L)$.

3. Circuits

Device (opt_in label)	Structure	Short arm (μm)	Long arm (μm)	ΔL (μm)	Expected FSR @1550 (nm)
opt_in_TE_1550_device_Javery_LB	GC $\rightarrow$ waveguide $\rightarrow$ GC loopback	—			— (GC envelope reference)
opt_in_TE_1550_device_Javery_MZI1	Y-branch MZI	128.4	179.4	51	11.2
opt_in_TE_1550_device_Javery_MZI2	Y-branch MZI	128.4	229.4	101	5.66
opt_in_TE_1550_device_Javery_MZI3	Y-branch MZI	128.4	329.4	201	2.84

Why three ΔL values. Over the 1499–1601 nm measurement sweep the MZIs show roughly 9, 18 and 36 fringes. The smallest ΔL keeps several fringes inside the GC passband even if the envelope is narrow; the largest ΔL gives the most fringes for an accurate FSR fit, while still having ~ 140 points per fringe at the 5000-point sweep. Fitting n_g from three independent circuits gives a consistency check and separates n_g from fabrication variations of ΔL . **Why a loopback.** The measured MZI spectrum is the product of two GC responses and the MZI response; the loopback measures the GC pair alone for baseline correction.

Components (SiEPIC EBeam PDK 0.4.53): GC_TE_1550_8degOxide_BB (black box, 127 μm pitch, input GC on top), ebeam_y_1550, waveguide type "Strip TE 1550 nm, w=500 nm" (bend radius 5 μm with Bezier 0.2).

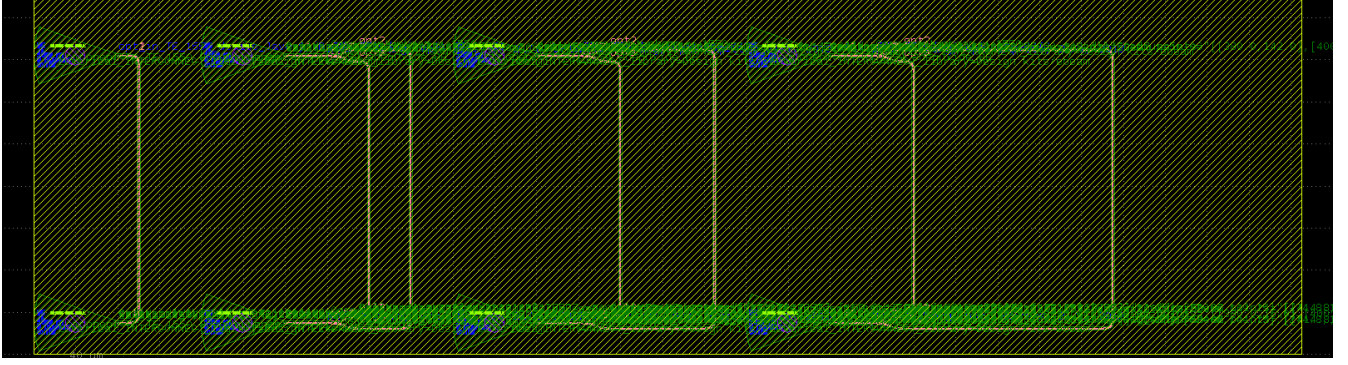


Fig. 1 – Layout, device region of the $605 \times 410 \mu\text{m}$ floorplan (upper part of the floorplan is empty). From left: loopback, MZI1, MZI2, MZI3. Each GC pair is $127 \mu\text{m}$ apart; the opt_in label sits on the upper (input) GC.

4. Expected spectra

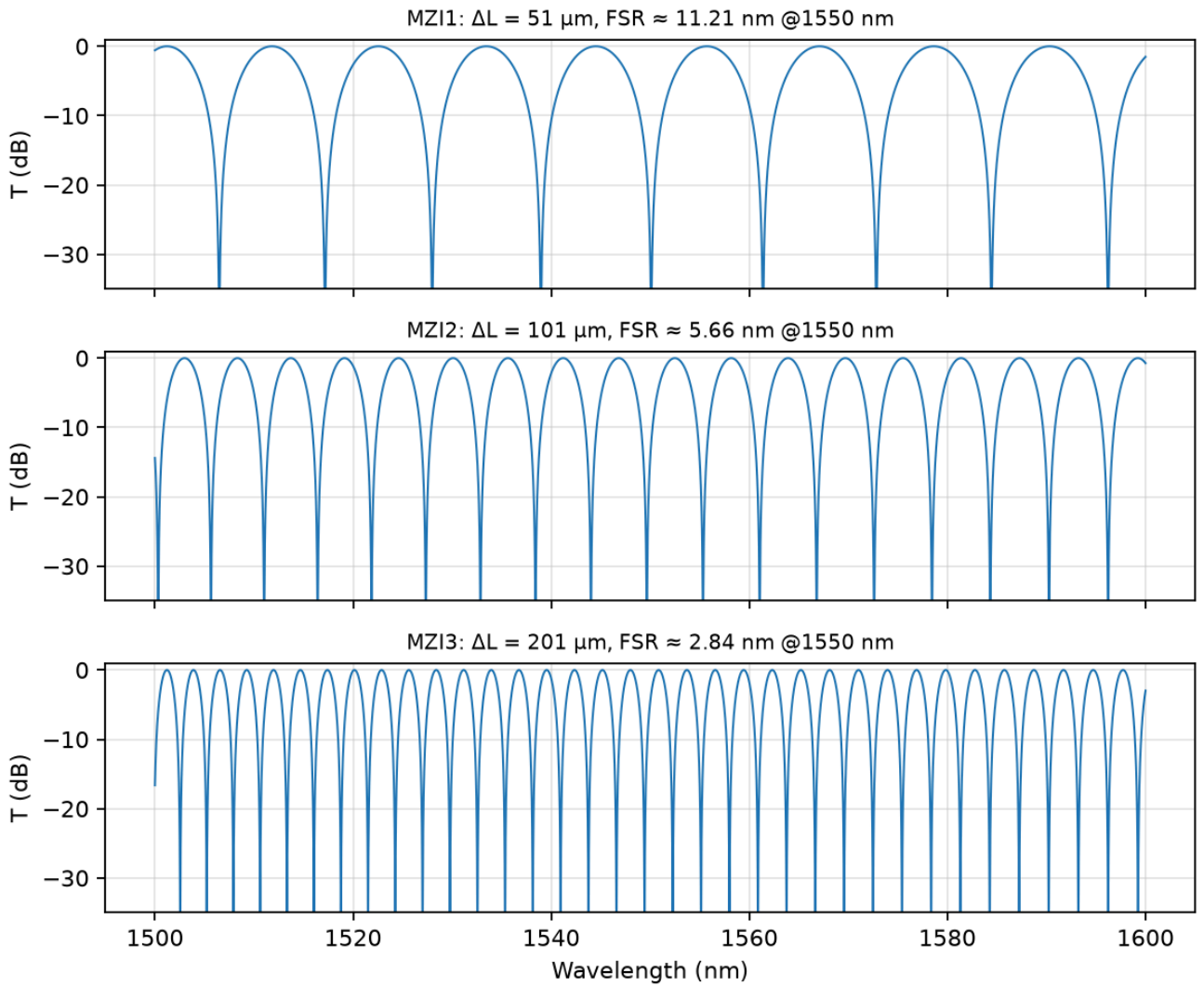


Fig. 2 – Ideal transmission of the three MZIs from the compact model (lossless Y-branches, GC envelope not included). The measured fringes will be modulated by the GC envelope seen in the loopback.

5. Supporting simulations

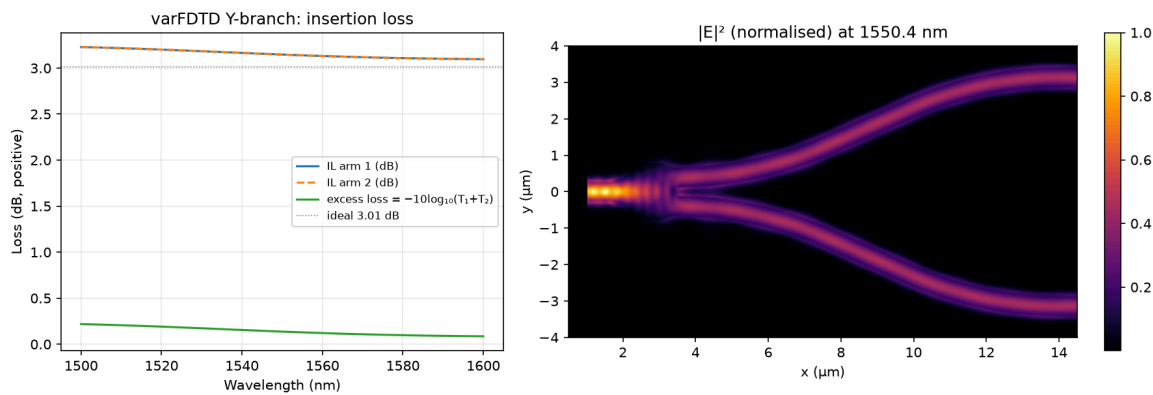


Fig. 3 – Y-branch (2.5D varFDTD, MODE): 3.15 dB per arm, excess loss 0.14 dB at 1550 nm (0.09–0.22 dB over 1500–1600 nm). Both arms equal → balanced MZI amplitudes, high extinction.

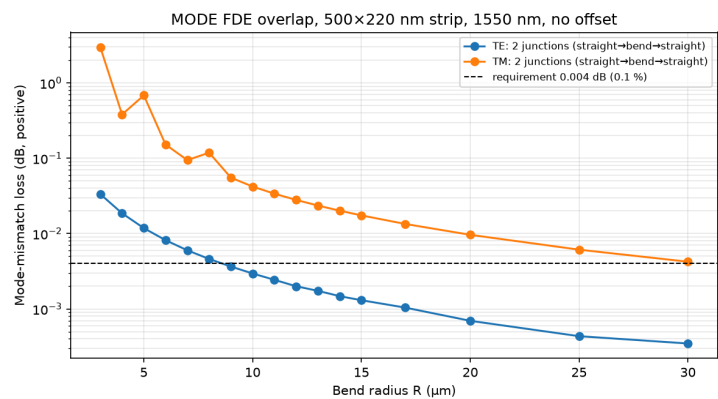


Fig. 4 – Mode-mismatch loss of a bend (straight→bend→straight, FDE overlap, TE/TM). At the PDK radius of 5 μm the TE loss is $\approx$ 0.012 dB per bend without the Bezier transition; the Bezier bends used in the layout reduce this further. The long arms contain only a few bends, so the arm-loss imbalance is small.

6. Verification

Check	Result
SiEPIC functional / DFT check (local, KLayout SiEPIC-Tools 0.5.31)	0 errors
Local PDK manufacturing DRC	0 items
Connectivity	4 independent circuits; unique opt_in labels; GCs at 127 μm pitch
openEBL-2026-10 GitHub checks (PR #248)	7/7 passed, merged 2026-10-03

7. Data analysis plan

(1) Normalise each MZI spectrum by the loopback spectrum. (2) Locate peaks/troughs (findpeaks) or use the autocorrelation of the spectrum to obtain $\text{FSR}(\lambda)$. (3) Compute $n_g(\lambda) = \lambda^2 / (\text{FSR} \cdot \Delta L)$ for each MZI and compare with the compact model (4.204 at 1550 nm). (4) Use the spread between the three circuits as an estimate of the extraction uncertainty. (*Simulations above are ideal / 2.5D / eigenmode models for design guidance, not fabrication sign-off.*)